

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

CSPIT/DEPSTAR

First Semester of B. Tech (EC,CE,IT,DCE,DCSE,DIT) Examination

November 2018

EE 141 Basics of Electronics and Electrical Engineering

Date: 20/11/2018, Tuesday

Time: 1:30 p.m. to 4:30 p.m.

Maximum Marks: 70

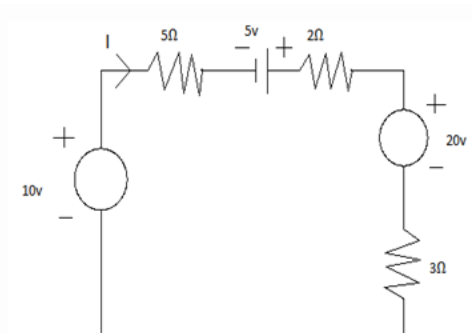
Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the questions below. [07]

- i. If $4\mu\text{F}$ and $2\mu\text{F}$ capacitors are connected in series, the equivalent capacitor is? [01]
(a) $2\mu\text{F}$ (b) $6\mu\text{F}$ (c) $0.75\mu\text{F}$ (d) $1.33\mu\text{F}$
- ii. The force per unit charge is known as? [01]
(a) Electric current (b) Electric field intensity (c) Electric potential (d) Electric Flux
- iii. When the current through the coil of an electromagnet reverses, the [01]
(a) magnetic field expands (b) direction of the magnetic field remains unchanged (c) magnetic field collapses (d) direction of the magnetic field reverses
- iv. The unit for permeability is [01]
(a) Wb (b) At/m (c) Wb/At $\times$ m (d) At/Wb
- v. Direction of induced current always [01]
(a) opposes the cause (b) remains same as that of cause (c) equal to cause that produces it (d) directs the cause
- vi. Solve for I in given fig. [02]
(a) -0.5A (b) 0.5A (c) -0.2A (d) 0.2A



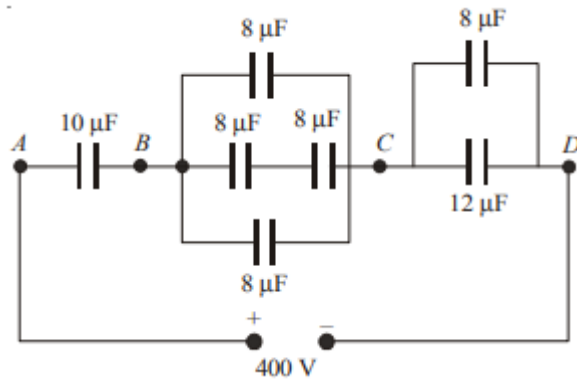
Q – 2. Answer any two questions. [14]

- A. (i.) A capacitor is composed of two plates separated by 3 mm dielectric of relative permittivity 4. [04]
An additional piece of insulation 5 mm thick is now inserted between the plates. If the capacitor has now capacitance one-third of its original capacitance, find the relative permittivity of the

additional dielectric.

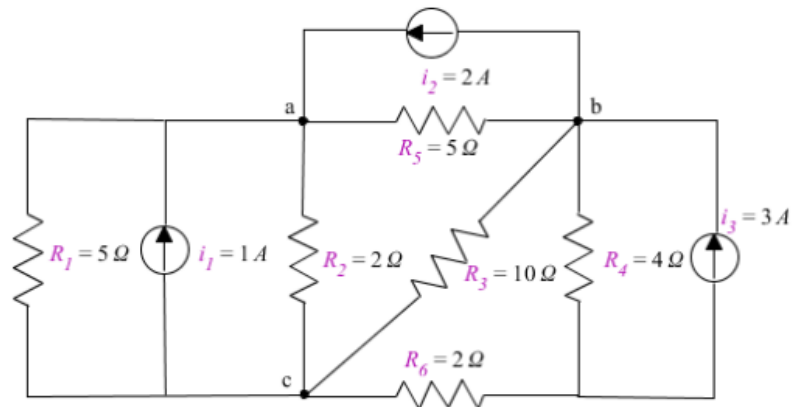
(ii.) In the circuit shown in Fig, find the equivalent capacitance between A and D

[03]

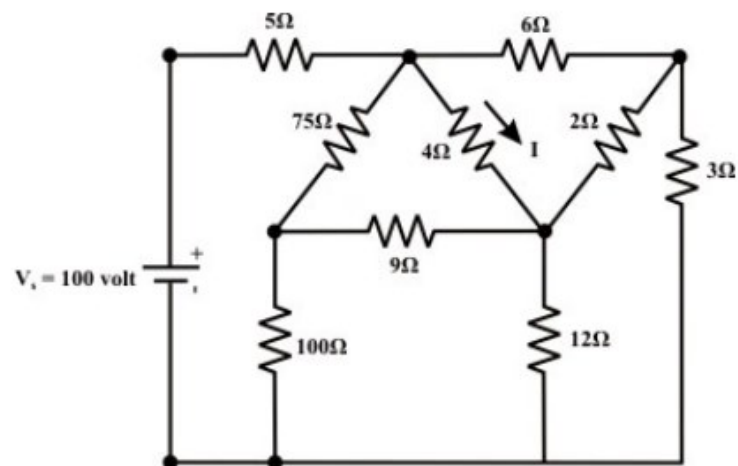


B. Find V_a , V_b , V_c using Nodal analysis

[07]

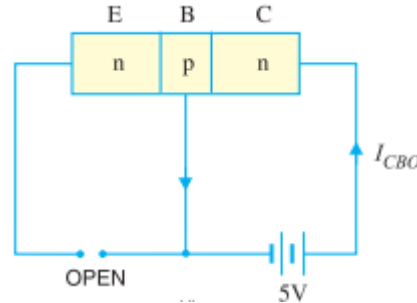


C. Find the current I through 4Ω resistor using $Y - \Delta$ or $\Delta - Y$ transformation technique only for the circuit shown in fig. [07]

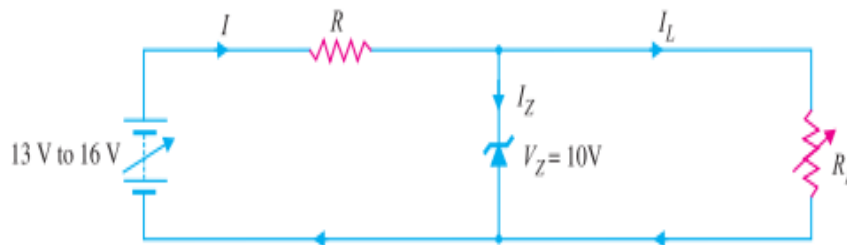


Q – 3. Answer any two questions.**[14]**

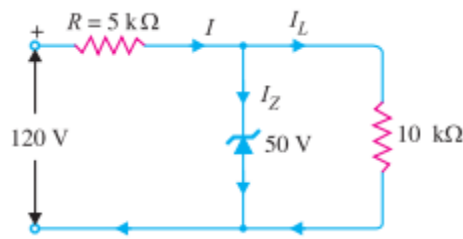
- A. An n-p-n transistor at room temperature has its emitter disconnected. A voltage of 5V is applied between collector and base. With collector positive, a current of $0.2 \mu\text{A}$ flows. When the base is disconnected and the same voltage is applied between collector and emitter, the current is found to be $20 \mu\text{A}$. Find α , I_E and I_B when collector current is 1mA . **[07]**



- B. (i.) A 10-V zener diode is used to regulate the voltage across a variable load resistor as shown in fig. The input voltage varies between 13 V and 16 V and the load current varies between 10 mA and 85 mA. The minimum zener current is 15 mA. Calculate the value of series resistance R. **[03]**



- (ii.) For the circuit shown in Fig, find: (a) the output voltage (b) the voltage drop across series resistance (c) the current through zener diode. **[04]**



- C. Derive the efficiency and PIV of full wave bridge rectifier. Draw necessary waveforms. **[07]**

SECTION – II

- Q - 4 Define following terms [07]**
 Waveform, Instantaneous value, Amplitude, Cycle, Time period, Frequency
 Alternation, PIV, Knee voltage
- Q-5 Answer any two questions [14]**
- A** (i.) The field winding of a generator has a resistance of 12.7Ω at 18°C and 14.3Ω at 50°C . find (a) temperature coefficient at 0°C , (b) resistance at 0°C , (c) temperature coefficient at 18°C [04]
- (ii.) Define a. Resistance b. Negative Temperature coefficient of resistance c. Positive Temperature coefficient of resistance [03]
- B** The flux of 1 mWb is produced by coil A when it carries a current of 4 A. Another coil B produces a flux of 1.5 mWb carrying the same current. Coil A has 800 turns and coil B has 1200 turns. The coils are kept such that 80% of flux produced by coil A links with coil B. Calculate a) Self-inductance of each coil b) Mutual inductance c) Co-efficient of coupling d) Percentage of flux produced by coil B that links with coil A [07]
- C** A series RLC circuit consist of $R = 1000 \text{ ohms}$, $L = 100 \text{ mH}$ and $C = 10\mu\text{F}$. if a voltage of 100 V is applied across the combination. [07]
- Find
- (a) the resonance frequency (b) Quality factor of the circuit at resonance frequency (c) current flowing in the circuit at resonance (d) Voltage across L and Voltage across C at resonance.
- Q-6 Answer any two questions [14]**
- a** Calculate form factor and peak factor of full wave rectified sine wave. [07]
- b** Explain power factor, active power, reactive power, apparent power, series R-L circuit with phasors diagram. [07]
- c** List the advantages of Three Phase system over single phase. [07]